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Machine learning-based identification of wastewater treatment plant-specific microbial indicators using 16S rRNA gene sequencing

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Effluent released from municipal wastewater treatment plants reflects the microbial communities responsible for degrading and removing contaminants within the plants. Monitoring this effluent offers essential insights into its environmental impacts, the efficiency of treatment processes, and the presence of emerging contaminants. To support improved monitoring and source attribution, our study employed a machine-learning framework to identify microbial indicators capable of distinguishing between municipal treatment plants based on effluent microbiota. We collected 57 effluent samples for sequencing of the V4 region of the 16S rRNA gene from six treatment plants in the Pirkanmaa region in Finland between 2016 and 2018. Characterising the microbiome revealed that although each plant had unique microbial profiles, their overall diversity and richness were similar. This provided a robust foundation for identifying plant-specific microbes. Using ANOVA-F for feature selection, we focused on the genus level due to its informative prevalence. Among various models tested, the Gaussian Naive Bayes model yielded the highest accuracy with the fewest relevant microbes. We identified nine bacterial genera and one archaeon, whose relative abundances predicted the origin of the effluent with 92% accuracy. Our study outlines a framework for the cost-effective and rapid identification of the origin of effluent or changes in the treatment process, demonstrating the power of machine learning in environmental monitoring and management.

Wastewater effluent is the treated water discharged from a wastewater treatment plant (WWTP) into surface waters or the environment. WWTPs significantly reduce bacterial loads, including pathogens and antibiotic-resistance genes. Thus, the effluent reflects the diversity and composition of microbial communities responsible for the breakdown and removal of organic matter, nutrients, and contaminants during the treatment process rather than the microbiome of the influent wastewater^{1–3}. To aid in the characterisation of these communities, the Microbial Database for Activated Sludge (MiDAS) compiles data from over 740 activated sludge plants and 280 anaerobic digesters, encompassing more than 100 000 species in its latest version (5.3)^{4,5}. Nevertheless, effluent microbiomes are dynamic, influenced by influent characteristics, seasonal variations, and treatment processes^{6,7}. Continuous surveillance of WWTP effluent is not only necessary for ensuring regulatory compliance but also for evaluating long-term ecological and environmental impacts by monitoring treatment efficiency⁸, detecting emerging contaminants^{9,10}, and tracking antibiotic resistance gene patterns^{3,11–15}. Additionally, wastewater microbiome profiling can support water source identification^{16–19}.

Faecal indicator bacteria such as *Escherichia coli* and *Enterococci* are commonly used to detect impact of WWTP effluent on receiving water bodies. However, they are not effective for monitoring treated effluent, as faecal bacteria are largely removed during the wastewater treatment process. Some faecally derived viruses, such as *Pepper mild mottle virus* and *crAssphage*, can survive treatment^{20,21}. Yet, these viral markers are too broadly distributed, and there are too few suitable viruses to differentiate between effluents from different WWTPs. Persistent chemical markers, including pharmaceuticals, solvents, plasticisers, antioxidants, and surfactants, are

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used to trace treated effluent in surface waters, and chemical fingerprinting has enabled the identification of specific industrial effluent sources^{22,23}. Still, there are no standardised methods to fingerprint effluents from individual municipal WWTPs.

Machine learning employs algorithms to create predictive models from input datasets, enabling quantitative and categorical predictions and identifying relevant features from extensive data sets. This robust analytic approach is particularly well-suited for wastewater microbiome analysis due to the large and complex nature of the data involved. While Principal Coordinate Analysis (PCoA) has long been used to interpret differences visually, applying comprehensive machine learning models to wastewater microbiomes remains relatively rare. Artificial neural networks have been utilised to predict the microbial compositions of activated sludge collected from WWTPs globally²⁴. Additionally, Kim et al.²⁵ applied support vector machines (SVM) and logistic regression to activated sludge microbiome data to predict the WWTPs' anaerobic, anoxic, and aerobic biological processes. Random Forest models have been employed to predict antibiotic-resistant gene profiles based on taxonomic composition, with model validation on hospital wastewater²⁶, and the Bayesian-based SourceTracker model has been used to trace antibiotic-resistant gene pollution sources¹⁷. Furthermore, Wang et al.²⁷ integrated metabolomic and metagenomic data, utilising multiple linear regression to predict microbial abundances from metabolite profiles. Random Forest and SVM models have also been used to identify membrane bioreactor-specific taxa from other WWTP systems¹⁹.

Our research aimed to identify a core group of microbial taxa in effluent that could reliably differentiate six closely situated municipal wastewater treatment plants (WWTPs), all discharging into a network of interconnected lakes and rivers. These plants use similar treatment processes, including activated sludge, which can make source differentiation challenging. By selecting a small set of discriminatory taxa, routine polymerase chain reaction (PCR)-based detection, such as quantitative or multiplex PCR, becomes feasible—still the quickest, cheapest, and most sensitive methods for targeted microbe identification. By employing ANOVA-F for feature selection and the Gaussian Naive Bayes model, we identified nine bacterial genera and one archaeon whose relative abundances could predict the specific WWTP with 92% accuracy. While the approach has not been field-validated, this study provides a proof of concept for microbial source tracking of municipal effluents. It illustrates how machine learning can support future development in environmental monitoring and wastewater management.

Material and methods

Sampling

We took effluent samples from six wastewater treatment plants in the Pirkanmaa region: WWTP1 (N=19), WWTP2 (N=10), WWTP3 (N=11), WWTP4 (N=5), WWTP5 (N=5) and WWTP6 (N=6). The WWTP1 plant serves 29 000 residents, WWTP2 1 500 residents, WWTP3 1 000 residents, WWTP4 200 000, WWTP5 2 900 and WWTP6 350 000 residents (Fig. 1). The study samples were collected from 2016 to 2018 and from effluent wastewater. Samples were taken as grab and composite samples, sometimes both on the same day. The samples were directly collected into 50 mL Falcon tubes at the collection sites. Composite samples were obtained using automatic samplers over 24 h. A 50 mL sample volume was immediately frozen at -20°C at the collection sites, and upon arrival at the laboratory, the samples were transferred to a -80°C freezer for storage until analysis.

DNA extraction

The samples were concentrated with Centricon Plus-70 3 K filters (Merck, Millipore) into a final volume of 700 μL and frozen at -80°C . DNA was extracted from the samples with DNeasy® PowerSoil® Kit (QIAGEN) according to the manufacturer's protocol. The extracted DNA was stored at -80°C .

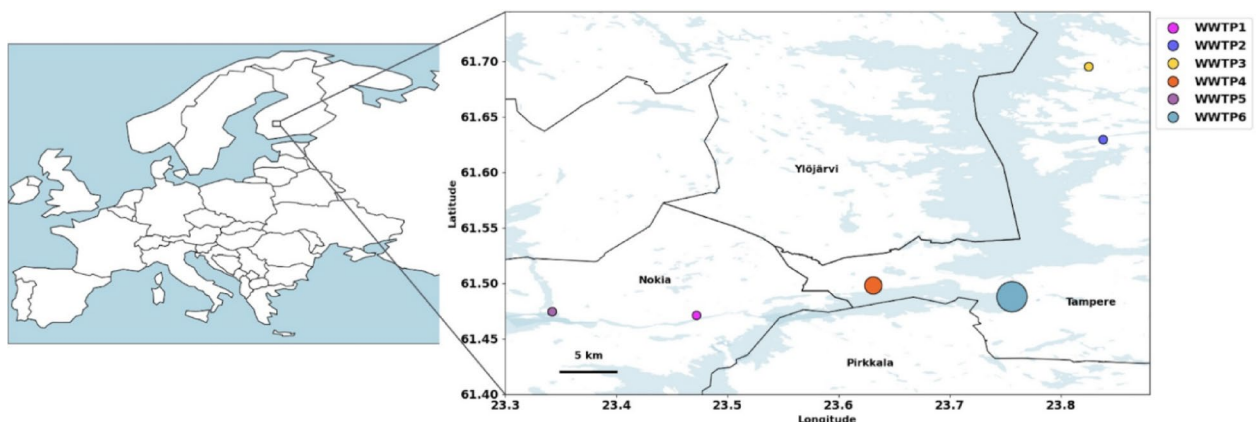


Fig. 1. We collected effluents from WWTPs in the Tampere region, Finland. The circles represent the plants' approximate locations, and each circle's size corresponds to the number of residents served by the WWTP. WWTP6 (light blue) and WWTP4 (orange) are the biggest and serve over 200,000 residents, whereas WWTP3 (yellow), WWTP5 (purple), WWTP2 (blue), and WWTP1 (pink) are small and serve between 1000 and 29,000 residents.

Amplification of 16S V4 rRNA gene, library preparation and sequencing

The microbiome library of all the samples was done by targeting the V4 region of the 16S rRNA gene. PCR was done using AccuPrime™ Pfx SuperMix (Invitrogen) with primers: 16Sf4V: AATGATACGCGACCGAC-CCGAGATCTACACTATGGTAATTGTGTGCCAGCMGCCGCGGTAA16Sr4VCAAGCAGAAGACGGGCAT-ACGAGATAGTCAGTCAGCCGGACTACHVGGGTWTCTAAT as described by Kozich et al.²⁸. The PCR reactions were purified with AMPure XP solution (Beckman Coulter Inc). The library DNA concentrations were quantified with a KAPA qPCR quantification kit (Illumina) and normalised to 2000 µM concentration. HT1 buffer (Illumina) was used to dilute the samples. Two microliters of the sample were used in the MiSeq-sequencing (Illumina) with PhiX control (Illumina). The read length was 250 nucleotides (paired-end, 2 × 250 bp). The raw reads are available from the NCBI SRA under BioProject ID PRJNA1177763, with accessions SAMN44450032–SAMN44450088.

Sequence analysis

QIIME2 (2024.2)²⁹ was used for raw data analysis. Raw reads were denoised and processed using the DADA2 plugin in QIIME 2, with a truncation length (p-trunc-len) of 150 bases. Chimeric sequences were identified and subsequently removed from the dataset, and the remaining sequences were then grouped into amplicon sequence variants (ASVs). The minimum number of ASVs per sample was 796, the maximum was 18 018, the median was 3030, and the mean was 4875.35. Taxonomy was assigned to ASVs using the q2-feature-classifier³⁰ and classify-sklearn naive Bayes taxonomy classifier against the MiDAS (5.3) database³¹. MiDAS includes provisional genus designations (e.g., midas_g_XXXX) for uncultured or poorly characterised taxa, and certain inconsistencies in naming conventions persist due to ongoing taxonomic revisions^{4,5}. ASVs were aligned with MAFFT³² via q2-alignment. Supplementary Table S1 contains the ASV table, including representative sequences, taxonomic assignments, and abundances across samples. Samples were rarefied (subsampling without replacement) to 950 sequences per sample before diversity analysis. Alpha diversity metrics, including Shannon's entropy³³ and Faith's Phylogenetic Diversity³⁴, and beta diversity based on Bray–Curtis distances followed by Principal Coordinate Analysis (PCoA), were calculated using the q2-diversity plugin in QIIME 2.

Machine learning, statistics, and visualisation

Statistical analyses and data visualisations were performed using Python libraries. Data preprocessing and manipulation were conducted with Pandas (2.2.1)³⁵ and NumPy (1.26.4)³⁹. Machine learning models and classification tasks were carried out using Scikit-learn (1.4.2)³⁸ and Imbalanced-learn (0.12.0)³⁶. Data visualization was performed with Matplotlib (3.8.4)⁴¹, Seaborn (0.13.2)³⁷, and Graphviz (9.0.0)⁴⁰. SHAP (0.44.1)⁴³ was used for model interpretability analyses, while statistical testing was done with SciPy (1.12.0)⁴² and Statsmodels (0.14.1)⁴⁴. Sample collection sites were mapped to provide geographic context for Fig. 1, using GeoPandas (0.14.3)⁴⁵, Matplotlib, and Shapely (2.0.3)⁴⁶ using map coordinates from Overpass API for OpenStreetMap⁴⁷. ANOVA-F feature selection was applied after testing for normality (Shapiro–Wilk) and homogeneity of variances (Levene's test), with 54.1% and 67.6% of taxa meeting assumptions, respectively. Random Forest, Gradient Boosting, Logistic Regression, Support Vector Machine, K-Nearest Neighbours, Naive Bayes, and Decision Tree models were trained using the selected genus-level features, with Synthetic Minority Oversampling Technique (SMOTE) applied to balance class distributions. Model evaluation was performed using five-fold cross-validation to estimate generalisation performance. In addition, bootstrap cross-validation and permutation testing were used to assess model stability and significance. Bootstrap resampling involved random sampling with replacement to calculate mean accuracy and standard deviation. Permutation testing was conducted by randomly shuffling the class labels to compute a null distribution of accuracies, providing a p-value for the observed model accuracy. All scripts, as well as the processed data frame containing relative abundances of bacterial and archaeal taxa (CodeEffsDataframe.csv), are available in the GitHub repository (https://github.com/jjurvans/Wastewater_Microbial_Markers). Initial versions of the scripts were generated with OpenAI's ChatGPT-4o⁴⁸.

Results

WWTP effluent bacterial phyla and diversity metrics

We collected effluent samples from six WWTPs discharging into interconnected surface waters in the Pirkanmaa region, Finland (Fig. 1). Constructed approximately 50 years ago, these plants serve similar populations due to their close geographical locations and utilise comparable treatment methods, primarily the activated sludge process for organic matter degradation. We found a total of 37 bacterial and archaeal phyla in the effluent samples. The dominant phyla based on relative abundance were *Proteobacteria*, *Bacteroidota*, and *Firmicutes* (Fig. 2a). In addition, WWTP1 and WWTP2 consistently exhibited higher proportions of *Actinobacteriota*, *Patescibacteria*, and *Campylobacterota*, whereas WWTP3, WWTP4, WWTP5, and WWTP6 displayed a more diverse composition of phyla. For example, in the largest plant (WWTP6), *Proteobacteria* dominated the effluent samples, accounting for 40–80% of the community, followed by *Bacteroidota* (5–20%) and *Firmicutes* (5–25%). *Patescibacteria* was the fourth most abundant phylum, contributing 2–15%. *Actinobacteriota*, *Campylobacterota*, *Verrucomicrobiota*, *Cyanobacteria*, and *Myxococcota* were present at lower abundances, each contributing less than 5% to the microbial community. To compare the WWTPs microbiome, we calculated the Bray–Curtis distance on rarefied samples, quantifying the difference between the two communities based on their ASV composition. Permutational multivariate analysis of sample variances (PERMANOVA) on Bray–Curtis distances showed that each WWTP microbiome differ significantly from the others (*P*-value < 0.006, *q*-value < 0.006) (Table 1). Principal Coordinate Analysis (PCoA) based on Bray–Curtis distance matrices was performed to assess differences in ASV profiles among the WWTP effluent samples (Fig. 2b). Samples from WWTP1 and WWTP3 formed partially distinct clusters, suggesting some separation between these plants. In

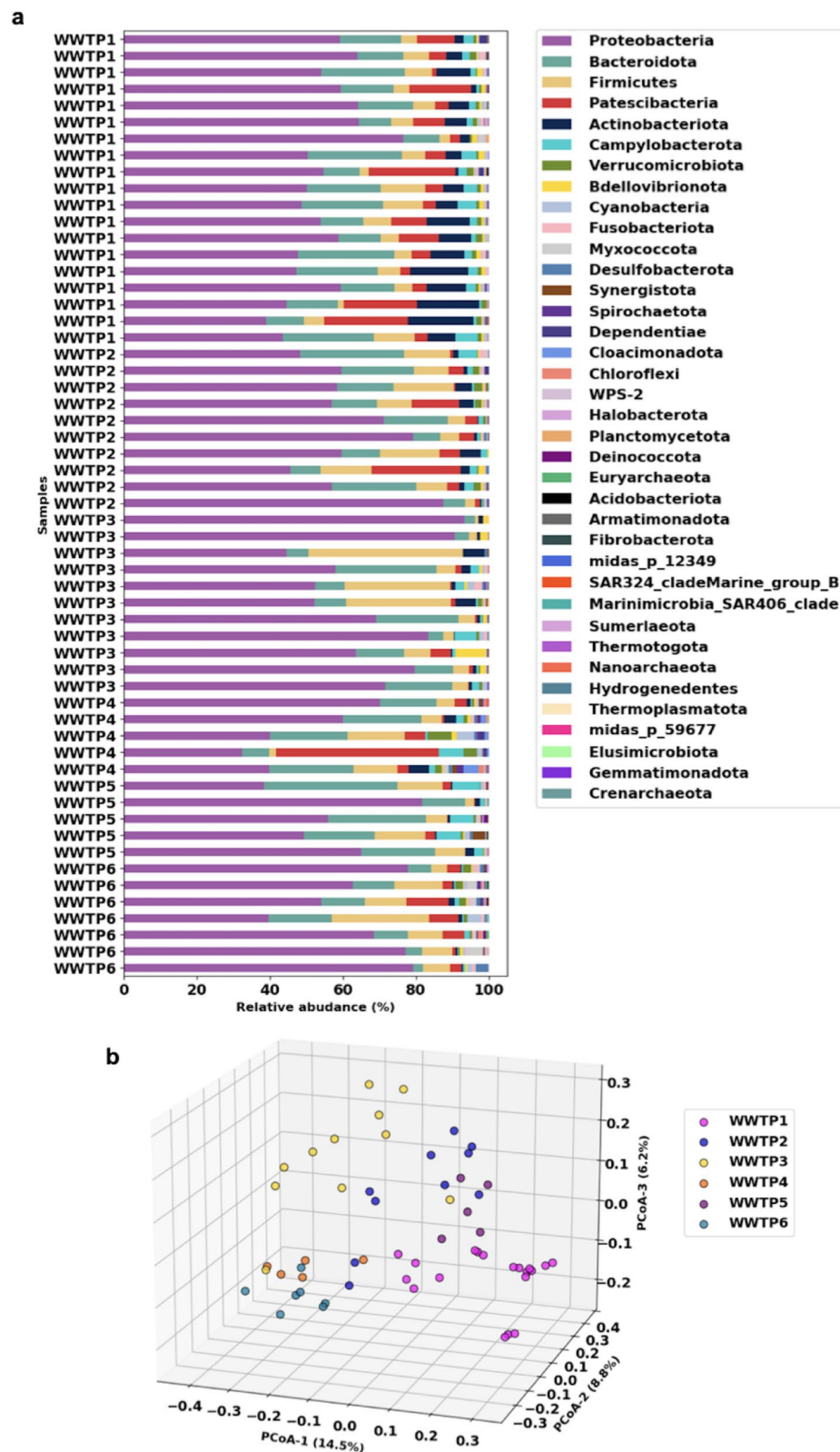


Fig. 2. Microbial community composition of the six Pirkanmaa region WWTP effluents. **(a)** Bar charts show the relative abundance (%) of bacterial phyla in individual samples (N = 57), grouped by WWTP. The y-axis indicates the WWTP for each group; multiple samples per WWTP are represented by individual bars without separate sample labels. A total of 37 bacterial and archaeal phyla were detected. **(b)** A principal coordinate analysis, based on a Bray–Curtis dissimilarity matrix, is depicted using tree axes, explaining 29.9% of the ASV variance.

Group 1	Group 2	Sample size	pseudo-F	<i>p</i> -value	<i>q</i> -value
WWTP1	WWTP3	30	6.949	0.001	0.0014
WWTP1	WWTP4	24	5.305	0.001	0.0014
WWTP1	WWTP5	24	2.673	0.002	0.00234
WWTP1	WWTP6	26	6.283	0.001	0.0014
WWTP2	WWTP1	29	4.429	0.001	0.0014
WWTP2	WWTP3	21	4.599	0.001	0.0014
WWTP2	WWTP4	15	3.784	0.001	0.0014
WWTP2	WWTP5	15	2.747	0.002	0.00234
WWTP2	WWTP6	17	4.095	0.001	0.0014
WWTP3	WWTP4	16	2.304	0.004	0.0043
WWTP3	WWTP5	16	2.925	0.001	0.0014
WWTP3	WWTP6	18	3.493	0.001	0.0014
WWTP4	WWTP5	10	3.267	0.006	0.006
WWTP4	WWTP6	12	2.347	0.001	0.0014
WWTP5	WWTP6	12	3.574	0.001	0.0014

Table 1. PERMANOVA with 999 permutations on Bray–Curtis distances to evaluate differences in the WWTPs’ microbiomes. The *Q*-value represents the *p*-value adjusted using the Benjamini & Hochberg correction.

contrast, samples from WWTP6 and WWTP4 clustered closely together, while WWTP2 and WWTP5 exhibited overlapping profiles. The first three PCoA axes explained 29.9% of the total variance in the data, indicating that only a limited proportion of the community variation is captured in three dimensions (PCoA1: 14.5%, PCoA2: 8.8% PCoA3: 6.2%). As a result, while some plant-specific clustering was observed, overall the PCoA could not fully separate the effluent microbiomes of all WWTPs.

We examined alpha diversity metrics on rarefied samples, including observed ASV richness, Faith’s phylogenetic diversity (Faith’s PD), and Shannon’s diversity index (Fig. 3). Differences between WWTPs were tested using the Mann–Whitney U test with Bonferroni correction for multiple comparisons (Table 2). Statistically significant differences across all three alpha diversity metrics were observed only between WWTP1 and WWTP2 ($P \leq 0.009$) and between WWTP1 and WWTP3 ($P \leq 0.002$). Specifically, WWTP1 exhibited higher microbial richness, with a mean observed ASV richness of 100, compared to WWTP2 (75) and WWTP3⁵¹. Faith’s PD values were 15 for WWTP1, compared to 11 for WWTP2 and 9 for WWTP3. Similarly, Shannon’s diversity index was higher in WWTP1 (5.7) than in WWTP2 (5.0) and WWTP3 (4.6). Overall, microbiome diversity and evenness appeared broadly similar across most WWTPs, as reflected by overlapping diversity metrics shown in Fig. 3.

Feature selection, model selection and model cross-validation

In feature and machine learning model selection, we aimed for a balance with the fewest features while maintaining model accuracy. We used ANOVA-F to calculate which bacteria and archaea have higher variance between WWTPs than inside the WWTPs in different taxa from phylum to genus. Due to the limited resolution of the V4 region, species-level taxonomic classification is considered unreliable and was not pursued in this study^{49,50}. We calculated the critical F-value at 95% confidence for each taxonomical level based on the number of bacterial or archaeal groups (features). The selected features had ANOVA-values higher than the critical F-value (Table 3). Genus-level features had the highest F-value (24.76) and the greatest number of selected features (263), compared to family and phylum levels (Table 3), and were therefore selected for model development.

We tested Random Forest, Gradient Boosting, Logistic Regression, Support Vector Machine, K-Nearest Neighbors, Naive Bayes, and Decision Tree models with the selected features. Synthetic Minority Over-Sampling Technique (SMOTE) was used to generate synthetic samples and balance the class distributions. We optimised the models with the application level of SMOTE from 10 to 100% and the number of selected features from 1 to 50. The models were assessed for accuracy, precision, recall, and the *f1_score* (Table 4). The Table 4 shows the results for the lowest feature counts and SMOTE application levels for the highest model metrics.

The best models were the Gaussian Naive Bayes classifier, with an accuracy score of 0.916, and the Random Forest, with an accuracy score of 0.833. The Gaussian Naive Bayes model consistently outperformed others across various metrics, utilising ten selected features and a SMOTE ratio of 0.4. With a SMOTE ratio of 0.4, the minor WWTP classes, WWTP5 and WWTP4, were augmented with two synthetic samples each, while WWTP6 received one synthetic sample. According to the classification report, WWTP2 and WWTP3 were the most challenging to classify (Table 5). These classification uncertainties were also evident in the Principal Component Analysis (PCA) with Gaussian Naive Bayes decision boundaries and uncertainty contours (See Supplementary Fig. S2 online).

The data was split into five folds for cross-validation, resulting in a mean cross-validation accuracy of 0.8, which is lower than the initial accuracy of 0.912. We also performed bootstrap cross-validation, yielding mean accuracy scores of 0.78 (std = 0.0953), and permutation cross-validation, yielding a model accuracy of 0.78 with a *P*-value of 0.001 and a mean permutation accuracy of 0.158 (std = 0.0573) (See Supplementary Fig. S1 online).

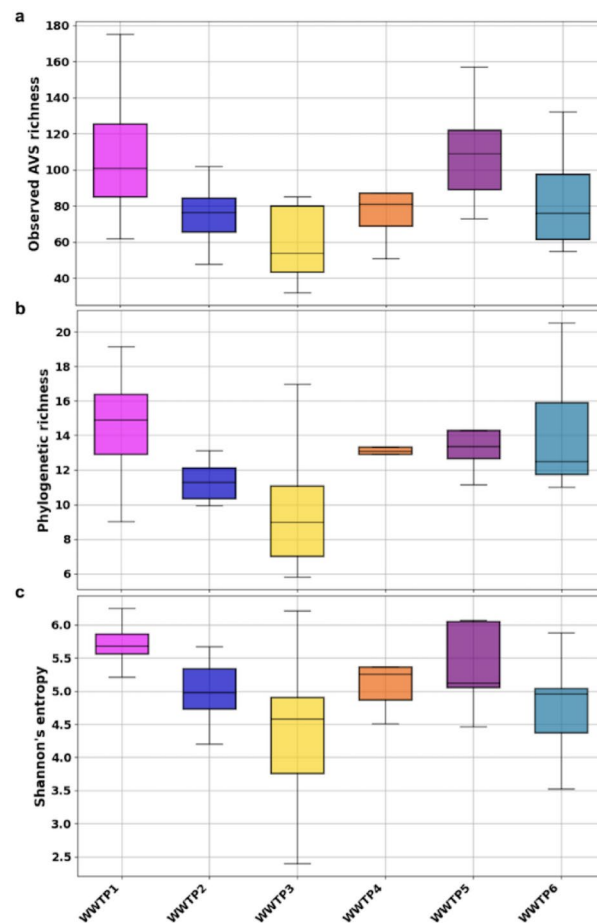


Fig. 3. Alpha-diversity metrics were calculated on rarefied sample reads: WWTP1 (N = 19), WWTP2 (N = 10), WWTP3 (N = 11), WWTP4 (N = 5), WWTP5 (N = 5) and WWTP6 (N = 7). The metrics include (a) Observed AVS richness, (b) Phylogenetic richness by Faith's phylogenetic diversity, and (c) Shannon's entropy estimation of richness and diversity.

Group 1	Group 2	P-value Observed richness	P-value Phylogenetic richness	P-value Shannon's entropy
WWTP1	WWTP2	0.009	0.001	0.001
WWTP1	WWTP3	0.002	0.000	0.000
WWTP1	WWTP4	0.227	0.489	0.160
WWTP1	WWTP5	0.831	0.446	0.446
WWTP1	WWTP6	0.083	0.364	0.007
WWTP2	WWTP3	0.149	0.038	0.170
WWTP2	WWTP4	0.806	0.028	0.594
WWTP2	WWTP5	0.028	0.019	0.206
WWTP2	WWTP6	0.845	0.070	0.887
WWTP3	WWTP4	0.145	0.013	0.069
WWTP3	WWTP5	0.019	0.013	0.145
WWTP3	WWTP6	0.135	0.003	0.425
WWTP4	WWTP5	0.222	1.000	1.000
WWTP4	WWTP6	0.876	0.876	0.343
WWTP5	WWTP6	0.202	0.876	0.149

Table 2. The Mann–Whitney U test with Benjamini & Hochberg correction assessed the differences in alpha diversity between the WWTPs.

Taxonomical level	Features	Critical F-value	Selected features	Highest F
Phylum	37	2.47	16	12.97
Class	87	2.32	41	12.19
Order	205	2.26	55	11.23
Family	422	2.24	135	12.19
Genus	891	2.22	263	24.76

Table 3. ANOVA-F-values and critical F-values were calculated for each feature from phylum to genus. The features with higher than the critical F-value are “Selected Features.” The highest F-value for each taxonomical level is shown.

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.833 (f = 35, SMOTE = 0.9)	0.816 (f = 47, SMOTE = 1.0)	0.833 (f = 35, SMOTE = 0.9)	0.791 (f = 35, SMOTE = 0.9)
Gradient Boosting	0.667 (f = 35, SMOTE = 0.0)	0.631 (f = 33, SMOTE = 0.0)	0.667 (f = 35, SMOTE = 0.0)	0.598 (f = 40, SMOTE = 0.0)
Logistic Regression	0.667 (f = 7, SMOTE = 0.0)	0.644 (f = 38, SMOTE = 0.0)	0.667 (f = 7, SMOTE = 0.0)	0.611 (f = 13, SMOTE = 0.0)
Support Vector Machine	0.583 (f = 7, SMOTE = 1.0)	0.694 (f = 36, SMOTE = 0.5)	0.583 (f = 7, SMOTE = 1.0)	0.605 (f = 36, SMOTE = 0.5)
K-Nearest Neighbors	0.583 (f = 7, SMOTE = 0.0)	0.766 (f = 36, SMOTE = 0.7)	0.583 (f = 7, SMOTE = 0.0)	0.619 (f = 36, SMOTE = 0.7)
Gaussian Naive Bayes	0.916 (f = 10, SMOTE = 0.4)	0.944 (f = 10, SMOTE = 0.4)	0.917 (f = 10, SMOTE = 0.4)	0.911 (f = 10, SMOTE = 0.4)
Decision Tree	0.750 (f = 10, SMOTE = 0.0)	0.600 (f = 10, SMOTE = 0.0)	0.750 (f = 10, SMOTE = 0.0)	0.657 (f = 10, SMOTE = 0.0)

Table 4. Performance metrics for machine learning model selection were optimised by minimising the number of selected features and adjusting the SMOTE ratio. The number of features is denoted by ‘f’. Bold values indicate the best-performing model for each metric.

WWTP	precision	recall	f1-score	support
WWTP1	1.000	1.000	1.000	4
WWTP2	1.000	0.500	0.667	2
WWTP3	0.667	1.000	0.800	2
WWTP4	1.000	1.000	1.000	1
WWTP5	1.000	1.000	1.000	1
WWTP6	1.000	1.000	1.000	2
accuracy			0.917	12
macro avg	0.944	0.917	0.911	12
weighted avg	0.944	0.917	0.911	12

Table 5. Classification report for Naive Bayes model with ten selected features and SMOTE ratio of 0.4

The drop in cross-validation accuracy scores was not statistically significant at the 95% confidence level, but it suggested potential overfitting to the original data fold. The accuracy of the unoptimised model on the entire dataset was 0.667. Thus, the feature and SMOTE-optimised model was more reliable than the baseline model and captured meaningful patterns in the data.

Characterisation of the relevant genera

The genera identified by the Naive Bayesian model to classify the six WWTPs in Pirkanmaa region are *Aquaspirillum arcticum* group, *midas_g_19012* (family *Microscillaceae*), *Pseudarcobacter*, DMER64 (family *Rikenellaceae*), ‘*Candidatus*’ *Cloacimonas* (family *Cloacimonadaceae*), *Sphingobium*, *Cloacibacterium*, *Smithella*, *OLB12* (family *Microscillaceae*), and archaea *Methanocorpusculum*. When plotting their relative abundances as percentages, we observed that their prevalence varies significantly between the WWTPs. In many cases, specific genera were present in only a few WWTPs, while absent in others, allowing effective differentiation between the WWTPs (Fig. 4a). Although these genera are relatively rare, they are not scarce: their mean abundance in WWTPs where they are detected typically ranges from 0.5 to 2%, except for *Methanocorpusculum*, which averages 0.08%. Beyond analysing relative abundances, we examined the contribution of each genus to WWTP classification using SHapley Additive exPlanations (SHAP) with the KernelExplainer method. Genera with

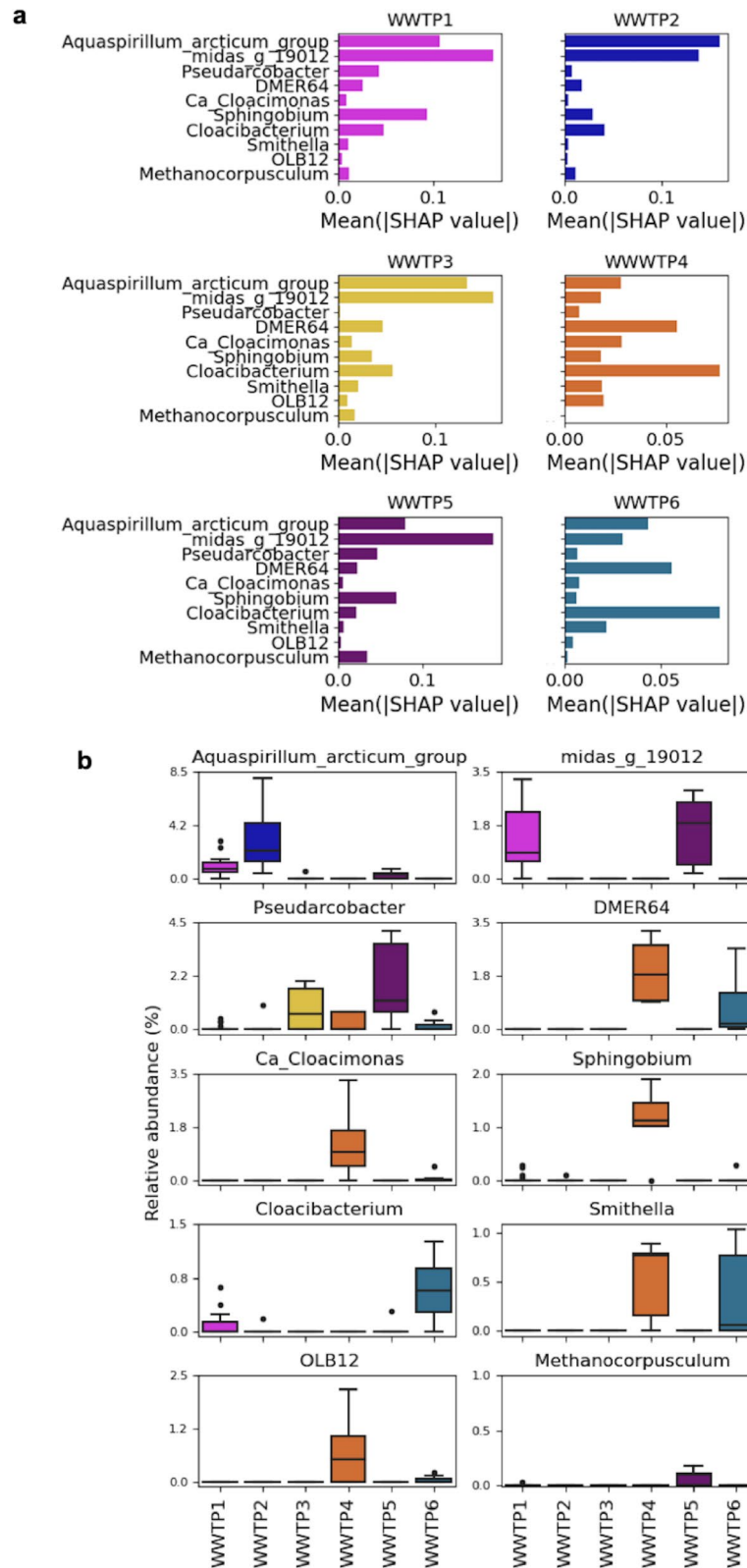


Fig. 4. The relevant bacteria and archaea that classify wastewater treatment plants using the Gaussian Naive Bayes model. **(a)** Relative abundance of the ten identified relevant bacteria and archaea (Methanocorpusculum) in the WWTP effluents. **(b)** SHAP (SHapley Additive exPlanations) summary plots for each WWTP. The SHAP values indicate the feature's average impact on the Gaussian Naive Bayes model's output magnitude. Each subplot corresponds to one WWTP and shows the mean absolute SHAP values. The higher the mean absolute SHAP values are, the more critical the microbe is for the model's predictions for the respective WWTP. Taxonomic assignment of uncultured taxa: midas_g_19012—Microscillaceae, DMER64—Rikenellaceae, Ca_Cloacimonas—Cloacimonadaceae, and OLB12—Microscillaceae.

Genera	Phylum	MiDAS classification
<i>Aquaspirillum arcticum</i> group	Proteobacteria	–
<i>Candidatus Cloacimonas</i>	Cloacimonadota	CRAT
<i>Cloacibacterium</i>	Bacteroidota	Loose core
DMER64 (<i>Rikenellaceae</i>)	Bacteroidota	CRAT
<i>Methanocorpusculum</i>	Halobacterota (Archaea)	–
<i>midas_g_19012</i> (<i>Microscillaceae</i>)	Bacteroidota	CRAT
OLB12 (<i>Microscillaceae</i>)	Bacteroidota	General core
<i>Pseudarcobacter</i>	Campylobacterota	Loose core
<i>Smithella</i>	Desulfobacterota	CRAT
<i>Sphingobium</i>	Proteobacteria	Loose core

Table 6. Phylum, MiDAS activated sludge prevalence classification for Gaussian Naive Bayes model relative genera. The MiDAS classification is explained in the text.

higher mean absolute SHAP values contributed more strongly to classification outcomes, independent of whether they increased or decreased prediction probability (Fig. 4b). Most genera influenced classification across multiple WWTPs, but each WWTP had two or three genera with the most substantial impacts. For example, the *Aquaspirillum* Arcticum group (a subgroup within the *Aquaspirillum* genus) and *midas_g_19012* (a provisional placeholder genus from the MiDAS database) were the primary contributors to WWTP2 classification: *Aquaspirillum* Arcticum group had a higher relative abundance in WWTP2 than in WWTP1 and WWTP5, while *midas_g_19012* was absent in WWTP2 but present in WWTP1 and WWTP5.

We consulted the MiDAS Field Guide and database publications, covering microbiomes from over 750 wastewater treatment plants^{4,5,51} to retrieve information on their prevalence and taxa (Table 6). This comparison assessed whether the genera differentiating the WWTPs in our study were widespread or site-specific. According to MiDAS, four of the identified genera belonged to the phylum *Bacteroidota*, two to *Proteobacteria*, and the remainder to *Campylobacterota*, *Desulfobacterota*, and *Cloacimonadota*. The genus *Methanocorpusculum* belongs to the archaeal phylum *Halobacterota*. All identified genera have been found in wastewater treatment plants, albeit with varied prevalence. The MiDAS database uses a standardised abundance-based categorisation: “conditionally rare or abundant” (CRAT) genera exceed 1% relative abundance in at least one plant, “loose core” genera exceed 0.1% abundance in more than 20% of plants, “general core” genera exceed 0.1% in more than 50% of plants, and “strict core” genera exceed 0.1% in more than 80% of plants. Based on these classifications, none of the genera identified in our study were strict core members. Four were classified as CRAT, three as loose core, and one (OLB12) as general core.

Discussion

Numerous studies have illustrated that the microbiome composition of wastewater varies significantly depending on its source, which includes different treatment processes, municipal wastewater, slaughterhouse effluents, refinery sewage, industrial discharges, and mine runoff^{16–19,52}. Furthermore, advanced machine learning models have been employed to predict the wastewater treatment processes in WWTPs based on the microbiome of activated sludge^{23,24}. Activated sludge is particularly interesting due to its diverse microbiome and critical role in wastewater treatment. Wijaya and Oh¹⁹ utilised a machine-learning approach to pinpoint distinctive microbial populations in membrane bioreactors, differentiating them from conventional activated sludge systems and sequencing batch reactors. Their research highlighted the genus *Ferruginibacter*, which exhibited metabolic activities well-suited to the operational conditions of membrane bioreactors.

Our objective was to pinpoint a core group of microbial markers capable of distinguishing WWTPs discharging into a shared surface water body, based on their effluent profiles. We focused on effluent samples because they could be used to reveal the identity of WWTPs from environmental discards. We examined the effluent microbiomes from six WWTPs that shared similarities in location, age, and treatment processes. Analysis of the relative abundances of microbial phyla and Bray–Curtis dissimilarity suggested that, while the microbiomes of these WWTPs were distinct, the phylogenetic structures and sequence richness within each plant were relatively consistent. To identify the minimal microbial markers predictive of WWTP origin, we selected microbes exhibiting the most significant variability between plants. We applied a machine-learning model to optimise and refine the selection. The Gaussian Naive Bayes model, which assumes normal distribution and independence among classes, proved the most accurate with the fewest microbial markers and minimal reliance on synthetic sample input. The assumption of normal distribution appeared reasonable given the relative abundance data of the microbes. However, it is important to note that microbial communities in activated sludge operate cooperatively as the microbiome self-assembles in response to environmental and operational conditions within bioreactors^{2,53}. Interestingly, the genera selected by the machine learning model were not part of the “strict core” microbiome observed in WWTPs globally, nor were they prevalent or abundant in the six WWTPs under study.

The nine bacteria and one archaea genera we identified as Pirkanmaa area WWTP markers were previously found in activated sludge or anaerobic digester^{4,5,51}. Among them, the genus *OLB12* (family *Microscillaceae*) stands out as the most prevalent globally, notably associated with ammonia removal through anaerobic oxidation^{4,54}. *Pseudarcobacter*, *Cloacibacterium*, and *Sphingobium* show significant global presence in over

20% of WWTPs, each exhibiting unique metabolic capabilities. *Pseudarcobacter* species are known for nitrate reduction and the hydrolysis of indoxyl acetate and urea⁵⁵. *Cloacibacterium* has demonstrated bioflocculation activity, aiding in removing heavy metals⁵⁶. The large genus *Sphingobium* is also found in contaminated soils and can degrade various organic compounds, including aromatic hydrocarbons, polycyclic aromatic hydrocarbons, polychlorophenols, and carbohydrates^{57,58}. Conversely, *DMER64* (family *Rikenellaceae*), *midas_g_19012* (family *Microscillaceae*), *Smithella*, and ‘Candidatus’ *Cloacimonas* are globally rare in WWTPs. *DMER64*, *Smithella* and ‘Candidatus’ *Cloacimonas* are associated with fatty acid degradation during methane production under varying conditions^{59–61}. In our data, *DMER64* and *Smithella* were identified concurrently in WWTP4 and WWTP6. The metabolic activity of *Midas_g_19012* has not been studied. *Aquaspirillum arcticum* group and *Methanocorpusculum* are among the rarest genera in activated sludge. *Aquaspirillum arcticum* was first isolated from Arctic sediment and is known to thrive at low temperatures, but its metabolic activity remains uncharacterised^{62,63}. *Methanocorpusculum* produces methane and is abundant in soil, animal gut, and as a cellular symbiont in marine ciliate^{64–66}. Thus, nine of these ten microbial markers play clear roles in degradation processes. Consequently, their specific abundances in WWTPs highlight the subtle variations in bioreactor microbiome self-assembly and the ecological conditions shaped by diverse environmental factors and reactor operations.

The machine-learning model demonstrates promising potential, yet its performance could be significantly enhanced by focusing on more hypervariable regions of the 16S rRNA gene or by employing shotgun metagenomic sequencing. This would enable precise species-level identification and functional analysis, respectively. The current model’s classification boundaries exhibit substantial uncertainties due to the limited and biased sample size. These uncertainties can be mitigated by sequencing a larger and more balanced number of samples from each WWTP. Nevertheless, dominant taxa are reliably captured, and the data span multiple years (2016–2018), reducing the likelihood of random patterns. Our findings present a proof of concept for identifying a core group of microbial taxa capable of distinguishing effluents from WWTPs with similar treatment processes. Though not yet validated for operational use, the approach highlights how microbial signatures—when combined with machine learning—can support the future development of source tracking tools for complex municipal wastewater networks, especially where multiple discharges enter connected surface waters.

Data availability

The datasets generated and/or analysed during the current study are available in the Supplementary Material (Supplementary Table S1), the GitHub repository (https://github.com/jjurvans/Wastewater_Microbial_Marker_s), and the NCBI SRA (<https://www.ncbi.nlm.nih.gov/sra/PRJNA1177763>).

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Declarations

Competing interests

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. Kirsi-Maarit Lehto, Heikki Hyöty and Sami Oikarinen are the stakeholders of GreenSeq Ltd. Finland.

Additional information

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